

Approach Note — Financial Document Q&A Chatbot

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Approach

I built a closed-book retrieval-augmented pipeline over 77 real SEC filings (10-Ks/10-Qs/8-Ks, ~38,800 chunks): PDF/HTML → page-aware chunking → local vector index → LLM answer generation with citations cross-checked against retrieved chunks, wrapped in a Flask app with drag-and-drop upload and a page-jump citation viewer. The one non-negotiable constraint behind every decision below: an answer that can't point to a specific document and page doesn't ship, even if it happens to be factually correct — a hallucinated-but-right answer counts as a failure, not a near-miss.

What I tried and measured

- **Document ingestion via Google Drive's API** — rejected after a controlled test showed real byte-level corruption risk: a base64 round-trip silently flipped one bit in a PDF's content stream, garbling the extracted text with no error and no broken citation. Replaced with byte-exact transfer only — drag-and-drop upload or CLI ingestion — never a text-relay path.
- **Retrieval backend** — dense embeddings (BGE + Chroma) were the original spec, but weren't installable in the build environment. I measured TF-IDF + SQLite/scipy as a substitute against the real corpus, confirmed it holds up on exact-figure and exact-phrase questions, and documented it as a swappable layer rather than a permanent choice.
- **A hard cosine-score threshold to gate “not found”** before calling the LLM — measured and rejected: a genuinely correct answer scored 0.13, while “what is the capital of France” scored 0.894 against an unrelated chunk dominated by the word “capital.” Score alone wasn't a reliable relevance signal, so I replaced the threshold with the LLM's own closed-book judgment as the sole not-found authority.
- **Conversational follow-up handling** — iterated three times against real multi-turn chains. Borrowing only the immediately-prior turn broke by question 3 of a chain; borrowing all prior turns unconditionally fixed that but let stale context bleed into a genuine topic change (a Nike→Pfizer switch stayed dominated by Nike terms). Final version borrows history only when the current question can't resolve a company on its own — a fresh, explicit mention always wins.
- **Real-question testing after the initial build** surfaced concrete retrieval gaps, each root-caused against the live index rather than assumed: CEO/officer lookups failing on vocabulary mismatch (“CEO” vs. “Chief Executive Officer”) and dilution inside long chunks; misspelled queries returning nothing, since TF-IDF only matches literal tokens; and fact-specific questions naming no company (e.g. a balance as of a specific date) failing to retrieve at all, because the only distinguishing signal — a date — had no boost mechanism to grab onto.

What I kept

TF-IDF/SQLite/scipy as the retrieval backend (zero marginal cost, holds up on this corpus, documented upgrade path); a forced tool-call answer schema with every citation cross-checked against what was actually retrieved (an unsupported citation downgrades the whole answer to not-found rather than being trusted); narrow, testable retrieval boosts layered on top of general TF-IDF — company-name-aware and date-aware — rather than a full rewrite; Haiku-by-default / Sonnet-for-cross-document-synthesis model routing to hit the cost target.

What I threw away

Google Drive as an ingestion source; the score-based not-found short-circuit; single-turn-only conversational context; a general-purpose Markdown parser for table rendering (kept it narrow — pipe tables and bold text only — so the fix couldn't become an HTML-injection surface).

Open

No golden test set sourced yet; near-tie conflicting-source resolution is implemented but untested against a real corpus conflict; the vector index isn't encrypted at rest (source documents are). Full detail on all of the above — including every measurement and rejected alternative — is in *docs/ARCHITECTURE.md* in the repository.